

Khush Purohit - 250840125028 Nikhil Lendkar - 250840125031

1. Draw line & bar charts to show urban and rural population of India over the years using population.csv [rural population is the difference between total population and urban population][For bar chart consider 8 yrs data]

```
In [17]: import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
```

```
In [18]: df = pd.read_csv("Population.csv")
df
```

Out[18]:

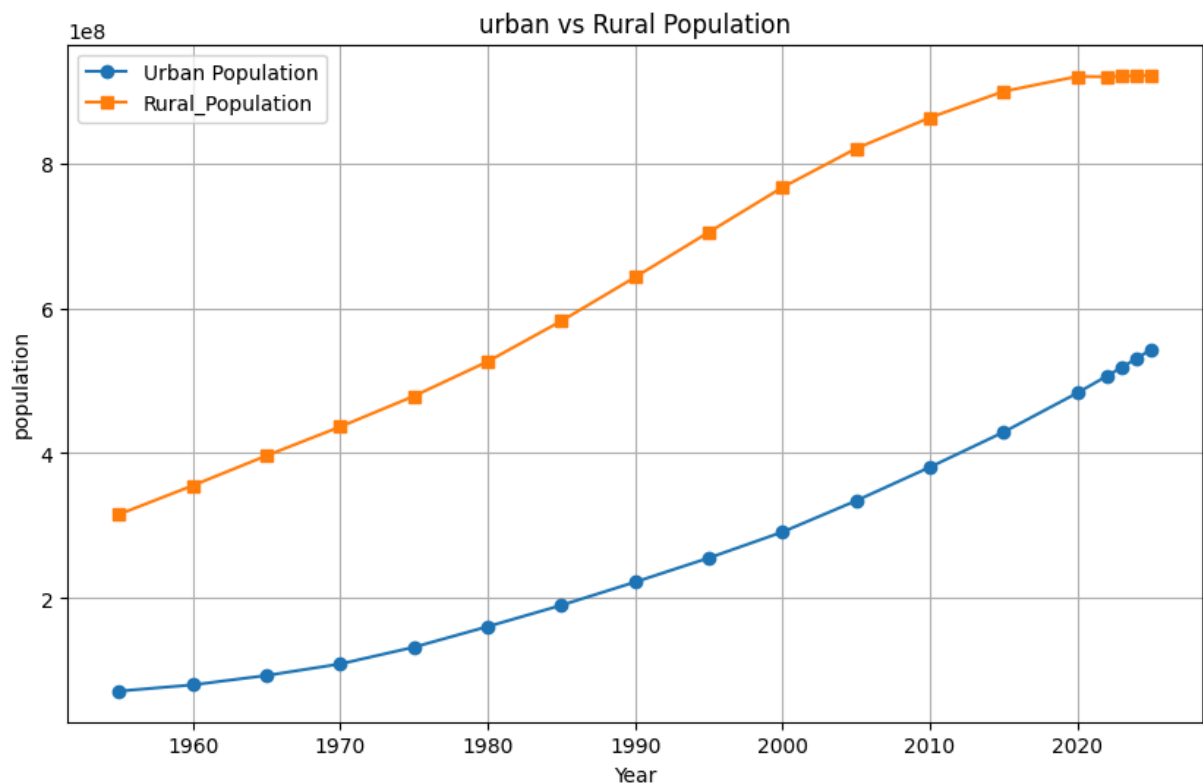
	Year	Population	Yearly % Change	Yearly Change	Median Age	Fertility Rate	Density	Urban Pop %	Urban Population
0	1955	387700887	2.29%	8284413	19.7	5.91	130	18.60%	71958495
1	1960	435990338	2.38%	9657890	19.2	5.92	147	18.50%	80565723
2	1965	490140146	2.37%	10829962	18.5	5.94	165	19.10%	93493844
3	1970	545864268	2.18%	11144824	18.1	5.62	184	20.00%	109388950
4	1975	611309535	2.29%	13089053	18.4	5.20	206	21.70%	132533810
5	1980	687354025	2.37%	15208898	18.9	4.78	231	23.40%	160941941
6	1985	772647793	2.37%	17058754	19.3	4.43	260	24.60%	190321782
7	1990	864972221	2.28%	18464886	19.7	4.04	291	25.70%	222296728
8	1995	960301044	2.11%	19065765	20.3	3.65	323	26.60%	255558824
9	2000	1057922733	1.96%	19524338	21.2	3.35	356	27.50%	291350282
10	2005	1154676322	1.77%	19350718	22.2	2.96	388	29.00%	334479406
11	2010	1243481564	1.49%	17761048	23.6	2.60	418	30.60%	380744554
12	2015	1328024498	1.32%	16908587	25.3	2.29	447	32.30%	429069459
13	2020	1402617695	1.10%	14918639	27.0	2.05	472	34.40%	483098640
14	2022	1425423212	0.81%	11402759	27.7	1.99	479	35.50%	506304869
15	2023	1438069596	0.89%	12646384	28.1	1.98	484	36.00%	518239122
16	2024	1450935791	0.89%	12866195	28.4	1.96	488	36.60%	530387142
17	2025	1463865525	0.89%	12929734	28.8	1.94	492	37.10%	542742539

```
In [ ]: df["Rural_Population"] = df["Population"] - df["Urban Population"]
```

```
In [20]: # plt.plot(rural,urb, label = "Urban pop. vs rural pop.")
# plt.xlabel("Urban population")
# plt.ylabel("Rural population")
# plt.legend()
# plt.show()

plt.figure(figsize=(10,6))
plt.plot(df["Year"], df["Urban Population"], label="Urban Population",marker= 'o')
plt.plot(df["Year"], df["Rural_Population"], label="Rural_Population",marker= 's')

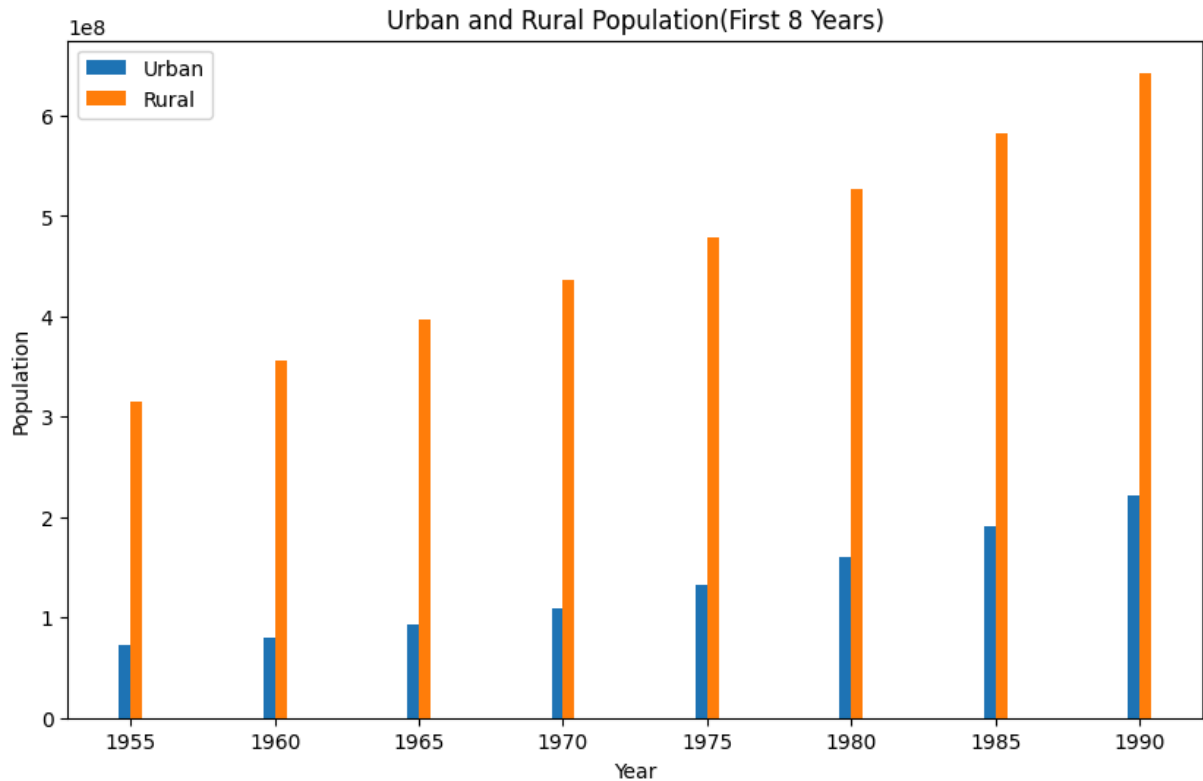
plt.title("urban vs Rural Population")
plt.xlabel("Year")
plt.ylabel("population")
plt.legend()
plt.grid(True)
plt.show()
```



```
In [ ]: subset = df.head(8)

plt.figure(figsize=(10,6))
plt.bar(subset["Year"] - 0.2, subset["Urban Population"], width=0.4, label="Urban")
plt.bar(subset["Year"] + 0.2, subset["Rural_Population"], width=0.4, label="Rural")

plt.title("Urban and Rural Population(First 8 Years)")
plt.xlabel("Year")
plt.ylabel("Population")
plt.legend()
plt.show()
```



2. Using gapminder.csv show life expectancy of top 10 highly populated countries from Asia

```
In [39]: df = pd.read_csv('gapminder2007.csv')
df
```

```
Out[39]:
```

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923.0	Asia	43.828	974.580338
1	Albania	3600523.0	Europe	76.423	5937.029526
2	Algeria	33333216.0	Africa	72.301	6223.367465
3	Angola	12420476.0	Africa	42.731	4797.231267
4	Argentina	40301927.0	Americas	75.320	12779.379640
...	...	...	...	...	...
137	Vietnam	85262356.0	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332.0	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743.0	Asia	62.698	2280.769906
140	Zambia	11746035.0	Africa	42.384	1271.211593
141	Zimbabwe	12311143.0	Africa	43.487	469.709298

142 rows × 5 columns

```
In [40]: # filtered = gap.sort_values(by='pop', ascending=False).head(10)
# filtered[['country', 'lifeExp']]

asia_df = df[df['continent'] == 'Asia']
```

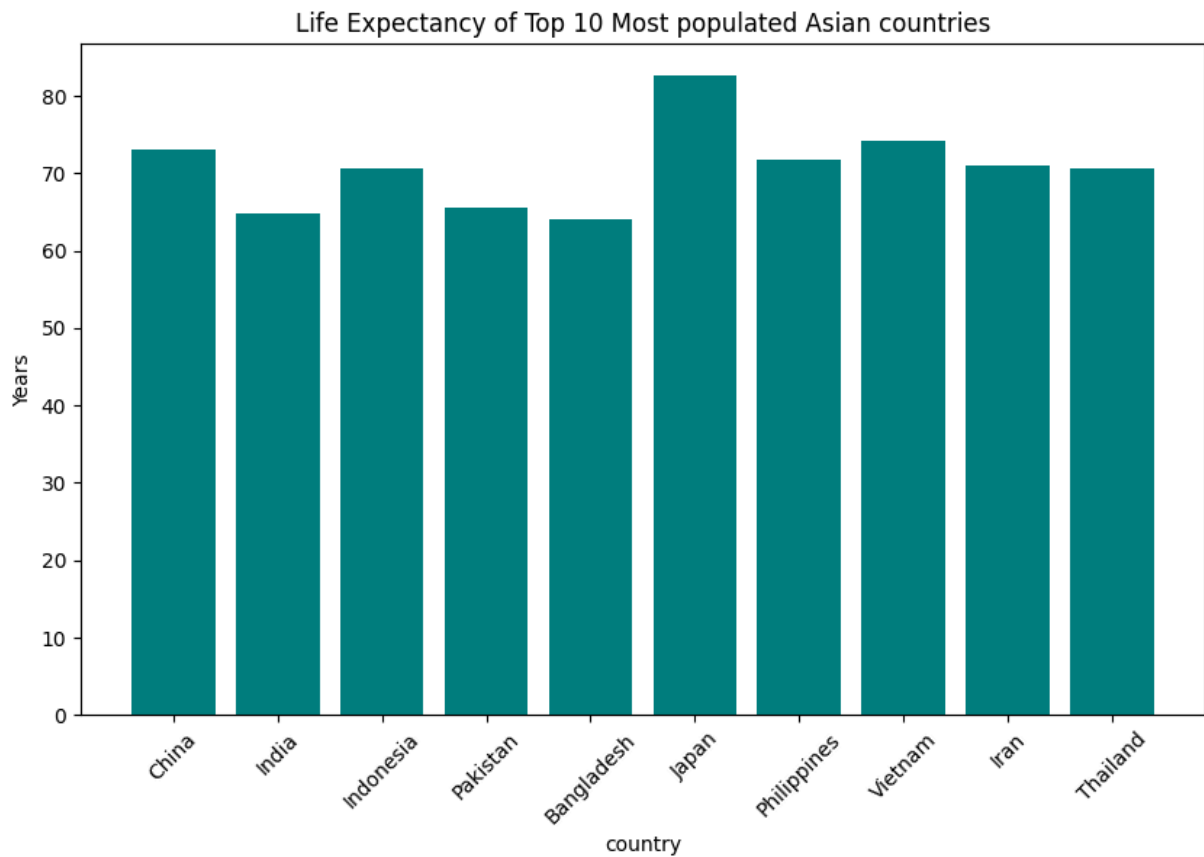
```
In [44]: top10 = asia_df.sort_values(by='pop', ascending=False).head(10)
```

```
In [45]: print(top10[['country', 'pop', 'lifeExp']])
```

	country	pop	lifeExp
24	China	1.318683e+09	72.961
58	India	1.110396e+09	64.698
59	Indonesia	2.235470e+08	70.650
97	Pakistan	1.692706e+08	65.483
8	Bangladesh	1.504483e+08	64.062
66	Japan	1.274680e+08	82.603
101	Philippines	9.107729e+07	71.688
137	Vietnam	8.526236e+07	74.249
60	Iran	6.945357e+07	70.964
127	Thailand	6.506815e+07	70.616

```
In [48]: plt.figure(figsize=(10,6))
plt.bar(top10['country'], top10['lifeExp'], color='teal')

plt.title("Life Expectancy of Top 10 Most populated Asian countries")
plt.xlabel("country")
plt.ylabel("Years")
plt.xticks(rotation=45)
plt.show()
```



3. Using a pie chart show distribution of top six selling cars [Create data as required]

```
In [49]: cars = ['Maruti Swift', 'Hyundai Creta', 'Tata Nexon', 'Mahindra XUV700', 'Toyota F
sales = [25000, 18000, 6500, 6000, 7500, 12000, 10000, 7000, 6500, 6000, 5500]

car = pd.DataFrame(
    {
        'carModel': cars,
        'sales': sales
    }
)
car
```

Out[49]:

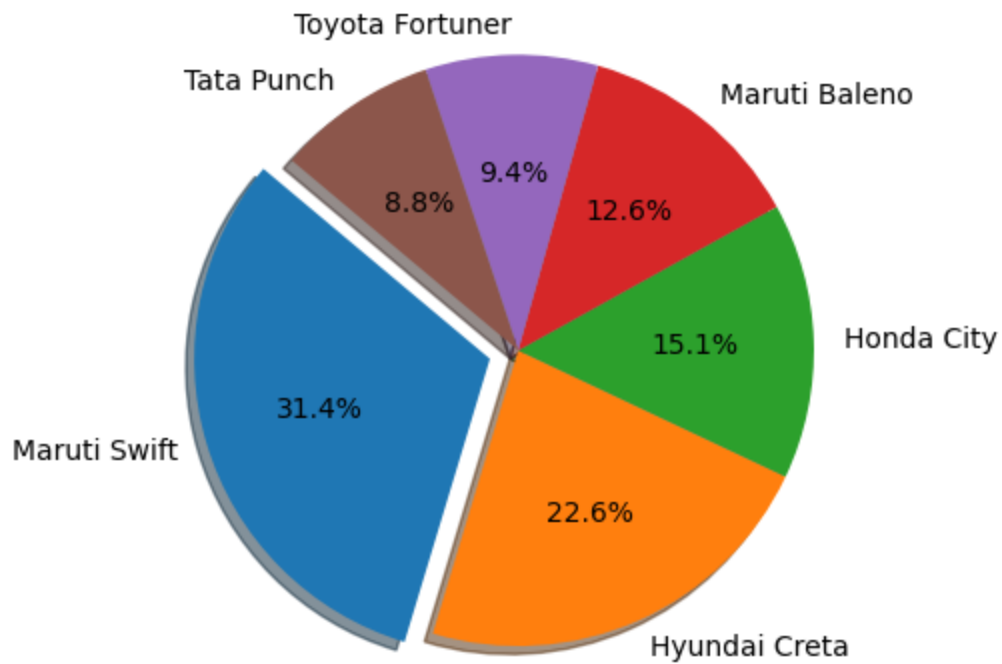
	carModel	sales
0	Maruti Swift	25000
1	Hyundai Creta	18000
2	Tata Nexon	6500
3	Mahindra XUV700	6000
4	Toyota Fortuner	7500
5	Honda City	12000
6	Maruti Baleno	10000
7	Tata Punch	7000
8	Hyundai Venue	6500
9	Kia Seltos	6000
10	MG Hector	5500

```
In [50]: car = car.sort_values(by='sales', ascending=False).head(6)
car
```

Out[50]:

	carModel	sales
0	Maruti Swift	25000
1	Hyundai Creta	18000
5	Honda City	12000
6	Maruti Baleno	10000
4	Toyota Fortuner	7500
7	Tata Punch	7000

```
In [54]: explode = [0.1, 0, 0, 0, 0, 0]
plt.pie(car['sales'], labels=car['carModel'], autopct='%1.1f%%', startangle=140, shadow=True)
plt.show()
```



4. Using Histogram show distribution of developers by their experience [use survey_data_sample.csv]

```
In [56]: sur = pd.read_csv('survey_results_sample.csv')
sur
```

```
Out[56]:
```

	ResponseId	Age	YearsCode	CompTotal	ConvertedCompYearly
0	1	18-24 years old	NaN	NaN	NaN
1	2	25-34 years old	18	2.850000e+05	285000.0
2	3	45-54 years old	27	2.500000e+05	250000.0
3	4	25-34 years old	12	1.560000e+05	156000.0
4	5	25-34 years old	6	1.320000e+06	23456.0
...	...	...	...	...	...
89179	89180	25-34 years old	20	2.000000e+05	NaN
89180	89181	18-24 years old	5	NaN	NaN
89181	89182	Prefer not to say	10	NaN	NaN
89182	89183	Under 18 years old	3	NaN	NaN
89183	89184	35-44 years old	17	3.300000e+09	NaN

89184 rows × 5 columns

In [59]: `sur.head(10)`

Out[59]:

	ResponseId	Age	YearsCode	CompTotal	ConvertedCompYearly
0	1	18-24 years old	NaN	NaN	NaN
1	2	25-34 years old	18	285000.0	285000.0
2	3	45-54 years old	27	250000.0	250000.0
3	4	25-34 years old	12	156000.0	156000.0
4	5	25-34 years old	6	1320000.0	23456.0
5	6	35-44 years old	21	78000.0	96828.0
6	7	35-44 years old	4	135000.0	135000.0
7	8	25-34 years old	5	80000.0	80000.0
8	9	45-54 years old	20	60000.0	64254.0
9	10	25-34 years old	6	NaN	NaN

In [60]: `experience = sur['YearsCode'].dropna()`

In []: `experience.isnull()`

Out[]:

```

1      False
2      False
3      False
4      False
5      False
...
89179  False
89180  False
89181  False
89182  False
89183  False
Name: YearsCode, Length: 87435, dtype: bool

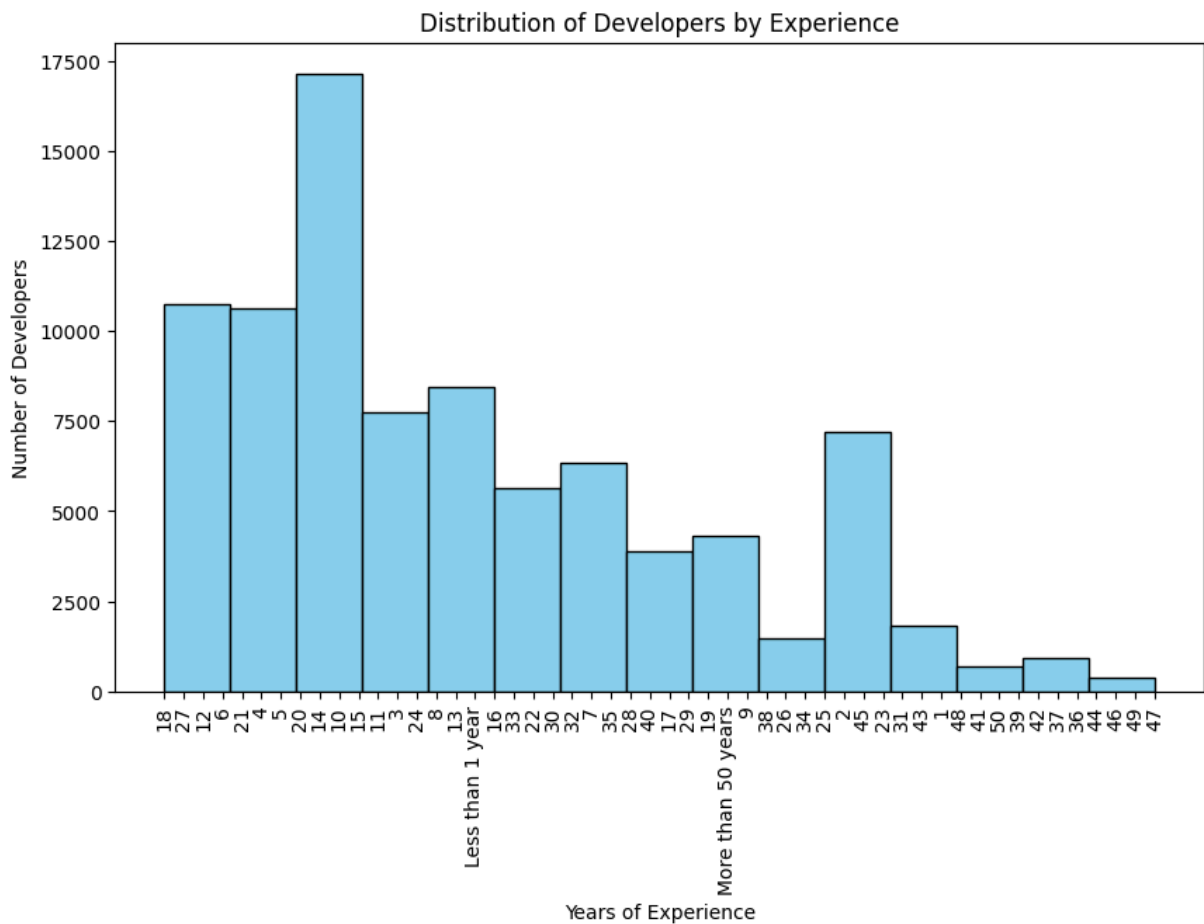
```

In [66]:

```

plt.figure(figsize=(10,6))
plt.hist(experience,bins=15,color='skyblue', edgecolor='black')
plt.title('Distribution of Developers by Experience')
plt.xlabel('Years of Experience')
plt.ylabel('Number of Developers')
plt.xticks(rotation=90)
plt.show()

```

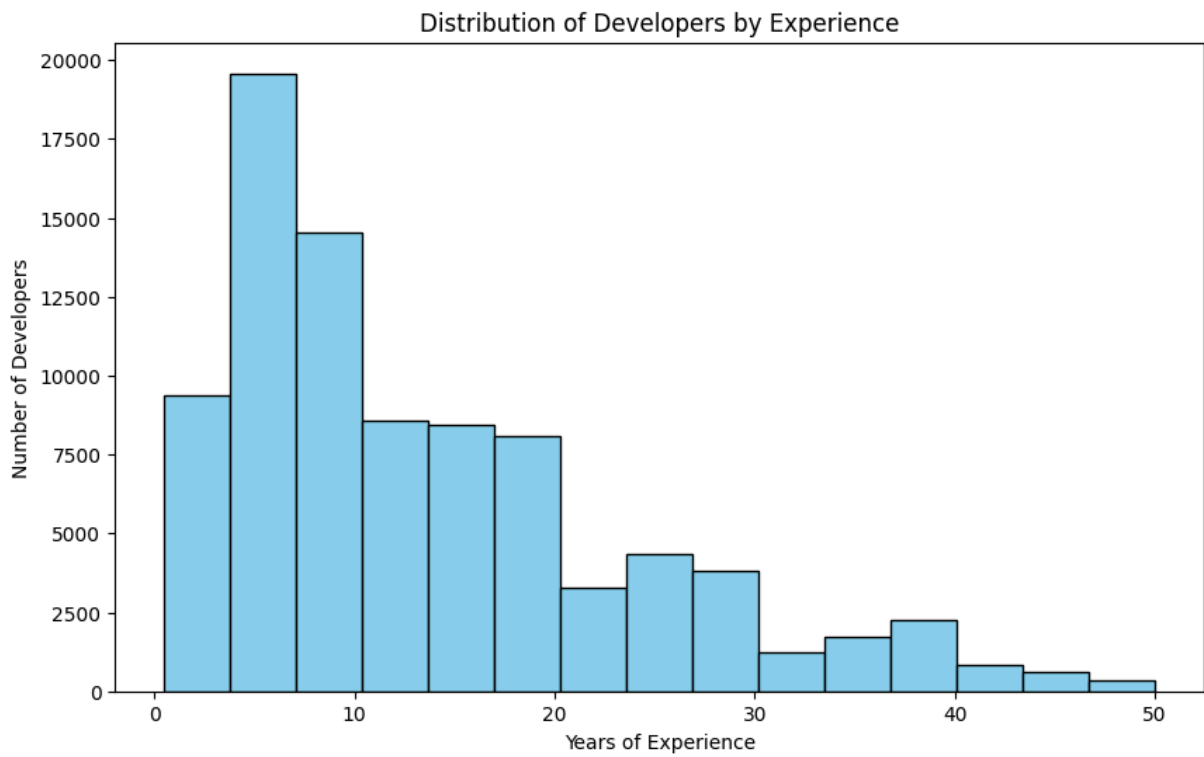



```
In [67]: experience = experience.replace({'Less than 1 year':0.5, 'More than 50 year':51})
experience = pd.to_numeric(experience,errors='coerce')
```

```
In [70]: experience.dropna()
```

```
Out[70]: 1      18.0
2      27.0
3      12.0
4       6.0
5      21.0
...
89179  20.0
89180   5.0
89181  10.0
89182   3.0
89183  17.0
Name: YearsCode, Length: 87082, dtype: float64
```

```
In [73]: plt.figure(figsize=(10,6))
plt.hist(experience,bins=15,color='skyblue', edgecolor='black')
plt.title('Distribution of Developers by Experience')
plt.xlabel('Years of Experience')
plt.ylabel('Number of Developers')
plt.show()
```



In []: